where

$$[\mathcal{S}] = \begin{bmatrix} 0 & -\omega_3 & 0 & v_1 \\ \omega_3 & 0 & 0 & v_2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

We can apply the matrix logarithm algorithm directly to $T_{sc}T_{sb}^{-1}$ to obtain $[\mathcal{S}]$ (and therefore $\mathcal{S}$) and θ as follows:

$$[\mathcal{S}] = \left[\begin{array}{ccc|c} 0 & -1 & 0 & 3.37 \\ 1 & 0 & 0 & -3.37 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right], \quad \mathcal{S} = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 3.37 \\ -3.37 \\ 0 \end{bmatrix}, \quad \theta = \frac{\pi}{6} \text{ rad (or } 30^\circ).$$

The value of $\mathcal{S}$ means that the constant screw axis, expressed in the fixed frame $\{s\}$, is represented by an angular velocity of 1 rad/s about $\hat{z}_s$ and a linear velocity (of a point currently at the origin of $\{s\}$) of $(3.37, -3.37, 0)$ expressed in the $\{s\}$ frame.

验证Schasles
-Mozzi定理

平面内任何位移
都可看作绕某点旋转

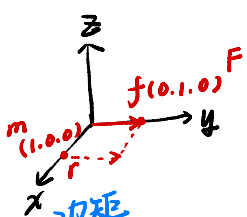
Alternatively, we can observe that the displacement is not a pure translation – T_{sb} and T_{sc} have rotation components that differ by an angle of 30° – and we quickly determine that $\theta = 30^\circ$ and $\omega_3 = 1$. We can also graphically determine the point $q = (q_x, q_y)$ in the $\hat{x}_s$–$\hat{y}_s$-plane through which the screw axis passes; for our example this point is given by $q = (3.37, 3.37)$.

For planar rigid-body motions such as this one, we could derive a planar matrix logarithm algorithm that maps elements of $SE(2)$ to elements of $se(2)$, which have the form

$$\begin{bmatrix} 0 & -\omega & v_1 \\ \omega & 0 & v_2 \\ 0 & 0 & 0 \end{bmatrix}.$$

3.4 Wrenches

$F = \begin{bmatrix} m \\ f \end{bmatrix} = \begin{bmatrix} r \times f \\ f \end{bmatrix}$ $\qquad \mathcal{V} = \begin{bmatrix} \omega \\ v \end{bmatrix}$



力矩

$m = r \times f = (0.0.1)$

沿z轴，大小1N·m

(让物体绕z轴旋转)

Consider a linear force f acting on a rigid body at a point r. Defining a reference frame $\{a\}$, the point r can be represented as $r_a \in \mathbb{R}^3$ and the force f can be represented as $f_a \in \mathbb{R}^3$. This force creates a **torque** or **moment** $m_a \in \mathbb{R}^3$ in the $\{a\}$ frame:

力矩

$$m_a = r_a \times f_a.$$

Note that the point of application of the force along its line of action is immaterial.

May 2017 preprint of *Modern Robotics*, Lynch and Park, Cambridge U. Press, 2017. http://modernrobotics.org

离原点

10 N	1 m	10N·m
10 N	0.5m	5N·m
5N	1 m	5N·m

⇒ 只有力矩 ⇏ 力大小/方向

只有力的大小方向 ⇏ 力矩

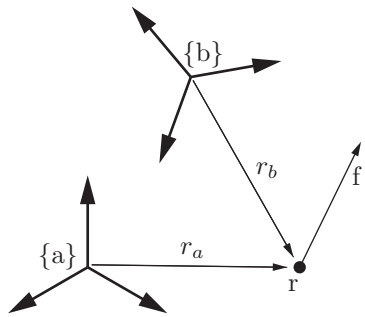


Figure 3.21: Relation between wrench representations $\mathcal{F}_a$ and $\mathcal{F}_b$.

Just as with twists, we can merge the moment and force into a single six-dimensional **spatial force**, or **wrench**, expressed in the {a} frame, $\mathcal{F}_a$:

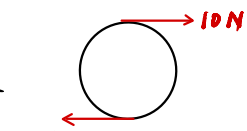
力旋量

$$\mathcal{F}_a = \begin{bmatrix} m_a \\ f_a \end{bmatrix} \in \mathbb{R}^6. \tag{3.93}$$

• 力偶 (Couple)
大小相等、方向相反
不在同条直线 的两个力

If more than one wrench acts on a rigid body, the total wrench on the body is simply the vector sum of the individual wrenches, provided that the wrenches are expressed in the same frame. A wrench with a zero linear component is called a **pure moment**. 纯力矩

f_a 线部分为0



f 合力 = 0
纯力矩：只有转动效果
无平动效果

A wrench in the {a} frame can be represented in another frame {b} (Figure 3.21) if T_{ba} is known. One way to derive the relationship between $\mathcal{F}_a$ and $\mathcal{F}_b$ is to derive the appropriate transformations between the individual force and moment vectors on the basis of techniques we have already used.

• wrench 的坐标变换

A simpler and more insightful way to derive the relationship between $\mathcal{F}_a$ and $\mathcal{F}_b$, however, is to (1) use the results we have already derived relating representations $\mathcal{V}_a$ and $\mathcal{V}_b$ of the same twist, and (2) use the fact that the power generated (or dissipated) by an $(\mathcal{F}, \mathcal{V})$ pair must be the same regardless of the frame in which it is represented. (Imagine if we could create power simply by changing our choice of reference frame!) Recall that the dot product of a force and a velocity is a power, and power is a coordinate-independent quantity. Because of this, we know that

功率 = 力 · 速度
= wrench 与 twist 点积

功率与坐标系无关

$$\mathcal{V}_b^{\mathrm{T}} \mathcal{F}_b = \mathcal{V}_a^{\mathrm{T}} \mathcal{F}_a. \tag{3.94}$$

From Proposition 3.22 we know that $\mathcal{V}_a = [\mathrm{Ad}_{T_{ab}}]\mathcal{V}_b$, and therefore Equation (3.94) can be rewritten as

$$\begin{aligned} \mathcal{V}_b^{\mathrm{T}} \mathcal{F}_b &= ([\mathrm{Ad}_{T_{ab}}]\mathcal{V}_b)^{\mathrm{T}} \mathcal{F}_a \\ &= \mathcal{V}_b^{\mathrm{T}} [\mathrm{Ad}_{T_{ab}}]^{\mathrm{T}} \mathcal{F}_a. \end{aligned}$$

力矩为0

在{h} $\mathcal{F}_h=(0,0,0,0,-5,0)$ } 都作用在对应坐标系原点（原点）

在{a} $\mathcal{F}_a=(0,0,0,0,0,1)$

给定了变换矩阵

T_{hf} T_{af}

$\mathcal{F}_f=[Ad_{T_{hf}}]^T\mathcal{F}_h+[Ad_{T_{af}}]^T\mathcal{F}_a$

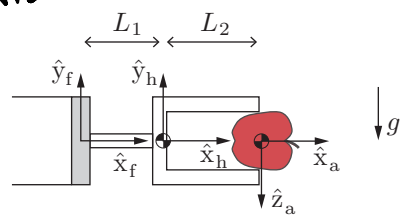


Figure 3.22: A robot hand holding an apple subject to gravity.

Since this must hold for all $\mathcal{V}_b$, this simplifies to

$$\mathcal{F}_b = [\mathrm{Ad}_{T_{ab}}]^{\mathrm{T}}\mathcal{F}_a. \tag{3.95}$$

wrench 坐标变换公式

Similarly,

$$\mathcal{F}_a = [\mathrm{Ad}_{T_{ba}}]^{\mathrm{T}}\mathcal{F}_b. \tag{3.96}$$

Proposition 3.27. *Given a wrench* F*, represented in* {a} *as* $\mathcal{F}_a$ *and in* {b} *as* $\mathcal{F}_b$*, the two representations are related by*

$$\mathcal{F}_b = \mathrm{Ad}^{\mathrm{T}}_{T_{ab}}(\mathcal{F}_a) = [\mathrm{Ad}_{T_{ab}}]^{\mathrm{T}}\mathcal{F}_a, \tag{3.97}$$

$$\mathcal{F}_a = \mathrm{Ad}^{\mathrm{T}}_{T_{ba}}(\mathcal{F}_b) = [\mathrm{Ad}_{T_{ba}}]^{\mathrm{T}}\mathcal{F}_b. \tag{3.98}$$

Since we usually have a fixed space frame {s} and a body frame {b}, we can define a **spatial wrench** $\mathcal{F}_s$ and a **body wrench** $\mathcal{F}_b$.

Example 3.28. The robot hand in Figure 3.22 is holding an apple with a mass of 0.1 kg in a gravitational field $g = 10$ m/s^2 (rounded to keep the numbers simple) acting downward on the page. The mass of the hand is 0.5 kg. What is the force and torque measured by the six-axis force–torque sensor between the hand and the robot arm?

We define frames {f} at the force–torque sensor, {h} at the center of mass of the hand, and {a} at the center of mass of the apple. According to the coordinate axes in Figure 3.22, the gravitational wrench on the hand in {h} is given by the column vector

$$\mathcal{F}_h = (0, 0, 0, 0, -5\ \mathrm{N}, 0)$$

and the gravitational wrench on the apple in {a} is

$$\mathcal{F}_a = (0, 0, 0, 0, 0, 1\ \mathrm{N}).$$

Given $L_1 = 10$ cm and $L_2 = 15$ cm, the transformation matrices T_{hf} and T_{af} are

只有平移,无旋转　　旋转+平移

$$T_{hf} = \begin{bmatrix} 1 & 0 & 0 & -0.1 \text{ m} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad T_{af} = \begin{bmatrix} 1 & 0 & 0 & -0.25 \text{ m} \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The wrench measured by the six-axis force–torque sensor is

$$\begin{aligned} \mathcal{F}_f &= [\mathrm{Ad}_{T_{hf}}]^{\mathrm{T}} \mathcal{F}_h + [\mathrm{Ad}_{T_{af}}]^{\mathrm{T}} \mathcal{F}_a \\ &= [0\ 0\ -0.5 \text{ Nm } 0\ -5 \text{ N } 0]^{\mathrm{T}} + [0\ 0\ -0.25 \text{ Nm } 0\ -1 \text{ N } 0]^{\mathrm{T}} \\ &= [0\ 0\ -0.75 \text{ Nm } 0\ -6 \text{ N } 0]^{\mathrm{T}}. \end{aligned}$$

3.5 Summary

The following table succinctly summarizes some of the key concepts from the chapter, as well as the parallelism between rotations and rigid-body motions. For more details, consult the appropriate section of the chapter.

Rotations	**Rigid-Body Motions**
$R \in SO(3) : 3 \times 3$ matrices $R^{\mathrm{T}}R = I, \det R = 1$	$T \in SE(3) : 4 \times 4$ matrices $T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$, where $R \in SO(3), p \in \mathbb{R}^3$
$R^{-1} = R^{\mathrm{T}}$	$T^{-1} = \begin{bmatrix} R^{\mathrm{T}} & -R^{\mathrm{T}}p \\ 0 & 1 \end{bmatrix}$
change of coordinate frame: $R_{ab}R_{bc} = R_{ac}, \quad R_{ab}p_b = p_a$	change of coordinate frame: $T_{ab}T_{bc} = T_{ac}, \quad T_{ab}p_b = p_a$
rotating a frame {b}: $R = \mathrm{Rot}(\hat{\omega}, \theta)$ $R_{sb'} = RR_{sb}$: rotate θ about $\hat{\omega}_s = \hat{\omega}$ $R_{sb''} = R_{sb}R$: rotate θ about $\hat{\omega}_b = \hat{\omega}$	displacing a frame {b}: $T = \begin{bmatrix} \mathrm{Rot}(\hat{\omega}, \theta) & p \\ 0 & 1 \end{bmatrix}$ $T_{sb'} = TT_{sb}$: rotate θ about $\hat{\omega}_s = \hat{\omega}$ (moves {b} origin), translate p in {s} $T_{sb''} = T_{sb}T$: translate p in {b}, rotate θ about $\hat{\omega}$ in new body frame
unit rotation axis is $\hat{\omega} \in \mathbb{R}^3$, where $\lVert\hat{\omega}\rVert = 1$	"unit" screw axis is $\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6$, where either (i) $\lVert\omega\rVert = 1$ or (ii) $\omega = 0$ and $\lVert v\rVert = 1$
	for a screw axis $\{q, \hat{s}, h\}$ with finite h, $\mathcal{S} = \begin{bmatrix} \omega \\ v \end{bmatrix} = \begin{bmatrix} \hat{s} \\ -\hat{s} \times q + h\hat{s} \end{bmatrix}$
angular velocity is $\omega = \hat{\omega}\dot{\theta}$	twist is $\mathcal{V} = \mathcal{S}\dot{\theta}$

continued...

Rotations (cont.)	**Rigid-Body Motions (cont.)**
for any 3-vector, e.g., $\omega \in \mathbb{R}^3$, $[\omega] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \in so(3)$ identities, $\omega, x \in \mathbb{R}^3, R \in SO(3)$: $[\omega] = -[\omega]^{\mathrm{T}}, [\omega]x = -[x]\omega$, $[\omega][x] = ([x][\omega])^{\mathrm{T}}, R[\omega]R^{\mathrm{T}} = [R\omega]$	for $\mathcal{V} = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6$, $[\mathcal{V}] = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix} \in se(3)$ (the pair (ω, v) can be a twist $\mathcal{V}$ or a "unit" screw axis $\mathcal{S}$, depending on the context)
$\dot{R}R^{-1} = [\omega_s], \quad R^{-1}\dot{R} = [\omega_b]$	$\dot{T}T^{-1} = [\mathcal{V}_s], \quad T^{-1}\dot{T} = [\mathcal{V}_b]$
	$[\mathrm{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6\times 6}$ identities: $[\mathrm{Ad}_T]^{-1} = [\mathrm{Ad}_{T^{-1}}]$, $[\mathrm{Ad}_{T_1}][\mathrm{Ad}_{T_2}] = [\mathrm{Ad}_{T_1T_2}]$
change of coordinate frame: $\hat{\omega}_a = R_{ab}\hat{\omega}_b, \quad \omega_a = R_{ab}\omega_b$	change of coordinate frame: $\mathcal{S}_a = [\mathrm{Ad}_{T_{ab}}]\mathcal{S}_b, \quad \mathcal{V}_a = [\mathrm{Ad}_{T_{ab}}]\mathcal{V}_b$
exp coords for $R \in SO(3)$: $\hat{\omega}\theta \in \mathbb{R}^3$	exp coords for $T \in SE(3)$: $\mathcal{S}\theta \in \mathbb{R}^6$
$\exp : [\hat{\omega}]\theta \in so(3) \to R \in SO(3)$ $R = \mathrm{Rot}(\hat{\omega}, \theta) = e^{[\hat{\omega}]\theta} =$ $I + \sin\theta[\hat{\omega}] + (1 - \cos\theta)[\hat{\omega}]^2$	$\exp : [\mathcal{S}]\theta \in se(3) \to T \in SE(3)$ $T = e^{[\mathcal{S}]\theta} = \begin{bmatrix} e^{[\omega]\theta} & * \\ 0 & 1 \end{bmatrix}$ where $* =$ $(I\theta + (1 - \cos\theta)[\omega] + (\theta - \sin\theta)[\omega]^2)v$
$\log : R \in SO(3) \to [\hat{\omega}]\theta \in so(3)$ algorithm in Section 3.2.3.3	$\log : T \in SE(3) \to [\mathcal{S}]\theta \in se(3)$ algorithm in Section 3.3.3.2
moment change of coord frame: $m_a = R_{ab}m_b$	wrench change of coord frame: $\mathcal{F}_a = (m_a, f_a) = [\mathrm{Ad}_{T_{ba}}]^{\mathrm{T}}\mathcal{F}_b$

3.6 Software

The following functions are included in the software distribution accompanying the book. The code below is in MATLAB format, but it is available in other languages. For more details on the software, consult the code and its documentation.

```
invR = RotInv(R)
```

Computes the inverse of the rotation matrix `R`.

`so3mat = VecToso3(omg)`
Returns the 3×3 skew-symmetric matrix corresponding to `omg`.

`omg = so3ToVec(so3mat)`
Returns the 3-vector corresponding to the 3×3 skew-symmetric matrix `so3mat`.

`[omghat,theta] = AxisAng3(expc3)`
Extracts the rotation axis $\hat{\omega}$ and the rotation amount θ from the 3-vector $\hat{\omega}\theta$ of exponential coordinates for rotation, `expc3`.

`R = MatrixExp3(so3mat)`
Computes the rotation matrix `R` $\in SO(3)$ corresponding to the matrix exponential of `so3mat` $\in so(3)$.

`so3mat = MatrixLog3(R)`
Computes the matrix logarithm `so3mat` $\in so(3)$ of the rotation matrix `R` $\in SO(3)$.

`T = RpToTrans(R,p)`
Builds the homogeneous transformation matrix `T` corresponding to a rotation matrix `R` $\in SO(3)$ and a position vector `p` $\in \mathbb{R}^3$.

`[R,p] = TransToRp(T)`
Extracts the rotation matrix and position vector from a homogeneous transformation matrix `T`.

`invT = TransInv(T)`
Computes the inverse of a homogeneous transformation matrix `T`.

`se3mat = VecTose3(V)`
Returns the $se(3)$ matrix corresponding to a 6-vector twist `V`.

`V = se3ToVec(se3mat)`
Returns the 6-vector twist corresponding to an $se(3)$ matrix `se3mat`.

`AdT = Adjoint(T)`
Computes the 6×6 adjoint representation $[\mathrm{Ad}_T]$ of the homogeneous transformation matrix `T`.

`S = ScrewToAxis(q,s,h)`
Returns a normalized screw axis representation $\mathcal{S}$ of a screw described by a unit

vector `s` in the direction of the screw axis, located at the point `q`, with pitch `h`.

`[S,theta] = AxisAng(expc6)`
Extracts the normalized screw axis $\mathcal{S}$ and the distance traveled along the screw θ from the 6-vector of exponential coordinates $\mathcal{S}\theta$.

`T = MatrixExp6(se3mat)`
Computes the homogeneous transformation matrix `T` $\in SE(3)$ corresponding to the matrix exponential of `se3mat` $\in se(3)$.

`se3mat = MatrixLog6(T)`
Computes the matrix logarithm `se3mat` $\in se(3)$ of the homogeneous transformation matrix `T` $\in SE(3)$.

3.7 Notes and References

The exponential coordinates for rotations introduced in this chapter are also referred to in the kinematics literature as the Euler–Rodrigues parameters. Other representations for rotations such as Euler angles, Cayley–Rodrigues parameters, and unit quaternions are described in Appendix B; further details on these and related parametrizations of the rotation group $SO(3)$ can be found in, e.g., [169, 113, 186, 122, 135].

Classical screw theory has its origins in the works of Mozzi and Chasles, who independently discovered that the motion of a rigid body can be obtained as a rotation about some axis followed by a translation about the same axis [25]. Ball's treatise [6] is often regarded as the classical reference on screw theory, while more modern treatments can be found in Bottema and Roth [18], Angeles [2], and McCarthy [113].

The identification of elements of classical screw theory with the Lie group structure of the rigid body motions $SE(3)$ was first made by Brockett in [20], who went considerably further and showed that the forward kinematics of open chains can be expressed as the product of matrix exponentials (this is the subject of the next chapter). Derivations of the formulas for the matrix exponentials, logarithms, their derivatives, and other related formulas can be found in [92, 129, 131, 122].